Project 7

Full Enterprise Network Simulation

Capstone Lab — Combining VLANs, Routing, GRE Tunneling, and ACLs

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1. Project Overview

This capstone project simulates a full enterprise network connecting a Headquarters (HQ) site with a remote Branch office. It integrates core networking technologies covered in Projects 1 through 6, including VLANs, inter-VLAN routing, static routing, GRE tunneling, and access control lists (ACLs).

Technologies Implemented

Technology	Purpose	Device
VLANs (10, 20, 30)	Department segmentation at HQ	Core-SW-HQ, Access-SW-HQ
802.1Q Trunking	Carry multi-VLAN traffic between switches	Core-SW-HQ, Access-SW-HQ
Inter-VLAN Routing (SVI)	Route traffic between VLANs	Core-SW-HQ (3560)
Static Routing	Route traffic between sites	Router-HQ, Router-Branch, Core-SW-HQ
GRE Tunnel (Tunnel10)	Encapsulate site-to-site traffic	Router-HQ, Router-Branch
Extended ACLs	Block Branch from HR and Management	Router-Branch

2. Network Topology

The network consists of two sites connected by a WAN link with a GRE tunnel overlay. HQ has three departments (IT, HR, Management) separated by VLANs, while the Branch office operates on a single flat subnet.

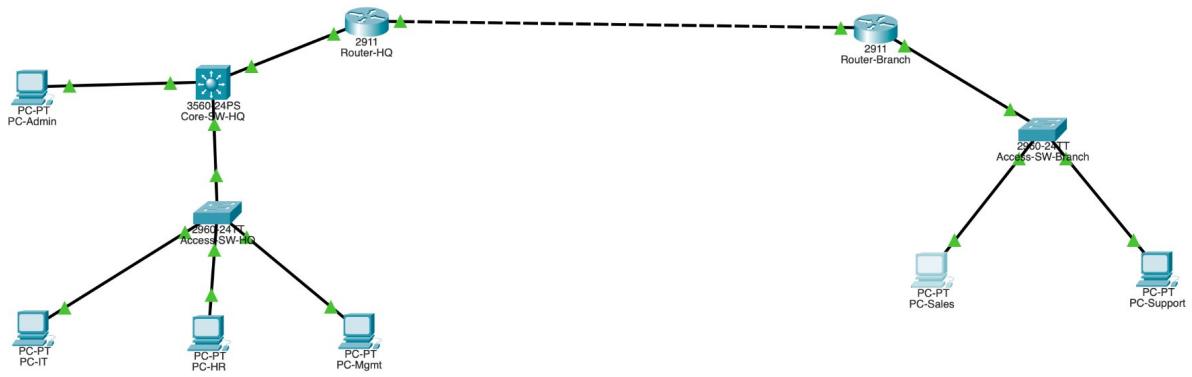

Figure 1: Full Enterprise Network Topology in Cisco Packet Tracer

Device Inventory

Device	Model	Role
Router-HQ	Cisco 2911	HQ gateway, GRE tunnel endpoint, WAN router
Router-Branch	Cisco 2911	Branch gateway, GRE tunnel endpoint
Core-SW-HQ	Cisco 3560-24PS	Layer 3 switch, inter-VLAN routing, SVI
Access-SW-HQ	Cisco 2960	Layer 2 access switch for HQ PCs
Access-SW-Branch	Cisco 2960	Layer 2 access switch for Branch PCs

3. IP Addressing Scheme

Device	Interface	IP Address	Subnet
Router-HQ	GigE0/0	192.168.1.1	255.255.255.0
Router-HQ	GigE0/1 (WAN)	10.0.0.1	255.255.255.252
Router-HQ	Tunnel10	172.16.0.1	255.255.255.252
Router-Branch	GigE0/0	192.168.40.1	255.255.255.0
Router-Branch	GigE0/1 (WAN)	10.0.0.2	255.255.255.252
Router-Branch	Tunnel10	172.16.0.2	255.255.255.252
Core-SW-HQ	GigE0/1	192.168.1.2	255.255.255.0
Core-SW-HQ	VLAN 10 (IT)	192.168.10.1	255.255.255.0
Core-SW-HQ	VLAN 20 (HR)	192.168.20.1	255.255.255.0
Core-SW-HQ	VLAN 30 (Mgmt)	192.168.30.1	255.255.255.0

End-Host Addressing

Host	IP Address	Gateway	VLAN/Subnet
PC-IT	192.168.10.10	192.168.10.1	VLAN 10 (IT)
PC-HR	192.168.20.10	192.168.20.1	VLAN 20 (HR)
PC-Mgmt	192.168.30.20	192.168.30.1	VLAN 30 (Mgmt)
PC-Admin	192.168.30.10	192.168.30.1	VLAN 30 (Mgmt)
PC-Sales	192.168.40.10	192.168.40.1	Branch LAN
PC-Support	192.168.40.20	192.168.40.1	Branch LAN

4. VLAN Configuration

Three VLANs were created on both Core-SW-HQ and Access-SW-HQ to segment HQ departments. An 802.1Q trunk on FastEthernet0/1 carries tagged VLAN traffic between the two switches.

VLAN ID	Name	Subnet	Purpose
10	IT	192.168.10.0/24	IT department workstations
20	HR	192.168.20.0/24	HR department workstations
30	Mgmt	192.168.30.0/24	Management and Admin PCs

VLAN Verification — show vlan brief

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/2
10   IT                               active
20   HR                               active
30   Mgmt                             active    Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Figure 2: VLANs 10, 20, 30 active on Core-SW-HQ

Trunk Verification — show interfaces trunk

```
Switch#show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       10,20,30

Port        Vlans allowed and active in management domain
Fa0/1       10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,20,30
```

Figure 3: 802.1Q trunk on Fa0/1 carrying VLANs 10, 20, 30

5. Inter-VLAN Routing (Layer 3 Switch)

Inter-VLAN routing is handled by the Core-SW-HQ (Cisco 3560) using Switch Virtual Interfaces (SVIs). Each VLAN has an SVI configured as the default gateway for hosts in that VLAN. The 'ip routing' command enables Layer 3 forwarding on the switch.

Core-SW-HQ SVI Configuration

```
interface Vlan10
ip address 192.168.10.1 255.255.255.0
interface Vlan20
ip address 192.168.20.1 255.255.255.0
interface Vlan30
ip address 192.168.30.1 255.255.255.0

interface GigabitEthernet0/1
no switchport
ip address 192.168.1.2 255.255.255.0
```

Core-SW-HQ Routing Table

```
Switch#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 192.168.1.1 to network 0.0.0.0

C    192.168.1.0/24 is directly connected, GigabitEthernet0/1
C    192.168.10.0/24 is directly connected, Vlan10
C    192.168.20.0/24 is directly connected, Vlan20
C    192.168.30.0/24 is directly connected, Vlan30
S*   0.0.0.0/0 [1/0] via 192.168.1.1
```

Figure 4: Core-SW-HQ routing table — connected VLANs + default route to Router-HQ

6. Static Routing

Static routes provide deterministic path selection across the network. Router-HQ acts as the central routing point, knowing the location of all subnets. Core-SW-HQ and Router-Branch each use a default route pointing to Router-HQ.

Router-HQ Static Routes

```
ip route 192.168.10.0 255.255.255.0 192.168.1.2 ! VLAN 10 via Core-SW
ip route 192.168.20.0 255.255.255.0 192.168.1.2 ! VLAN 20 via Core-SW
ip route 192.168.30.0 255.255.255.0 192.168.1.2 ! VLAN 30 via Core-SW
ip route 192.168.40.0 255.255.255.0 172.16.0.2 ! Branch via GRE Tunnel
```

```
Router-HQ#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
     172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/30 is directly connected, Tunnel10
L       172.16.0.1/32 is directly connected, Tunnel10
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
S    192.168.10.0/24 [1/0] via 192.168.1.2
S    192.168.20.0/24 [1/0] via 192.168.1.2
S    192.168.30.0/24 [1/0] via 192.168.1.2
S    192.168.40.0/24 [1/0] via 172.16.0.2
```

Figure 5: Router-HQ routing table — static routes to all subnets

Router-Branch Static Routes

```
ip route 0.0.0.0 0.0.0.0 172.16.0.1 ! Default route via GRE Tunnel
```

```
Router-Branch#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 172.16.0.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
S       10.0.0.1/32 is directly connected, GigabitEthernet0/1
L       10.0.0.2/32 is directly connected, GigabitEthernet0/1
     172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.0.0/30 is directly connected, Tunnel10
L       172.16.0.2/32 is directly connected, Tunnel10
     192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/0
L       192.168.40.1/32 is directly connected, GigabitEthernet0/0
S*   0.0.0.0/0 [1/0] via 172.16.0.1
```

Figure 6: Router-Branch routing table — default route through tunnel

Core-SW-HQ Default Route

```
ip route 0.0.0.0 0.0.0.0 192.168.1.1 ! Default route to Router-HQ
```

7. GRE Tunnel Configuration

A Generic Routing Encapsulation (GRE) tunnel (Tunnel10) is established between Router-HQ and Router-Branch over the WAN link. GRE encapsulates routed traffic inside a point-to-point tunnel, providing a logical direct connection between the two sites. All inter-site traffic is routed through this tunnel using the 172.16.0.0/30 overlay network.

Router-HQ Tunnel10 Configuration

```
interface Tunnel10
ip address 172.16.0.1 255.255.255.252
tunnel source GigabitEthernet0/1
tunnel destination 10.0.0.2
```

```
Router-HQ#show interface tunnel10
Tunnel10 is up, line protocol is up (connected)
  Hardware is Tunnel
  Internet address is 172.16.0.1/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 10.0.0.1 (GigabitEthernet0/1), destination 10.0.0.2
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     19 packets input, 1908 bytes, 0 no buffer
     Received 4 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 input packets with dribble condition detected
     0 packets output, 0 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
      0 unknown protocol drops
     0 output buffer failures, 0 output buffers swapped out
```

Figure 7: Router-HQ Tunnel10 — up/up, GRE/IP encapsulation

Router-Branch Tunnel10 Configuration

```
interface Tunnel10
ip address 172.16.0.2 255.255.255.252
tunnel source GigabitEthernet0/1
tunnel destination 10.0.0.1
```

```
Router-Branch#show interface tunnel10
Tunnel10 is up, line protocol is up (connected)
  Hardware is Tunnel
  Internet address is 172.16.0.2/30
  MTU 17916 bytes, BW 100 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel source 10.0.0.2 (GigabitEthernet0/1), destination 10.0.0.1
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 1
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     15 packets input, 1720 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 input packets with dribble condition detected
     0 packets output, 0 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
      0 unknown protocol drops
     0 output buffer failures, 0 output buffers swapped out
```

Figure 8: Router-Branch Tunnel10 — up/up, GRE/IP encapsulation

8. Access Control Lists (ACLs)

An extended ACL (101) is applied inbound on Router-Branch's GigE0/0 interface to enforce security policy. Branch users (Sales and Support) are blocked from accessing the HR (VLAN 20) and Management (VLAN 30) networks, but are permitted to reach IT (VLAN 10) for support requests.

ACL Configuration on Router-Branch

```
access-list 101 deny ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255
access-list 101 deny ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255
access-list 101 permit ip any any

interface GigabitEthernet0/0
ip access-group 101 in
```

Security Policy Summary

Source	Destination	Action	Reason
Branch (192.168.40.0)	HR (192.168.20.0)	DENY	Sensitive HR data
Branch (192.168.40.0)	Mgmt (192.168.30.0)	DENY	Admin-only access
Branch (192.168.40.0)	IT (192.168.10.0)	PERMIT	Tech support requests
Any	Any	PERMIT	All other traffic allowed

ACL Verification — show access-lists

```
Router-Branch#show access-lists
Extended IP access list 101
    10 deny ip 192.168.40.0 0.0.0.255 192.168.20.0 0.0.0.255 (4 match(es))
    20 deny ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255 (4 match(es))
    30 permit ip any any (11 match(es))
```

Figure 9: ACL 101 with match counters — deny rules actively blocking traffic

9. Verification & Testing

Test 1: Inter-VLAN Routing — PC-IT to PC-HR

PC-IT (VLAN 10, 192.168.10.10) successfully pings PC-HR (VLAN 20, 192.168.20.10), confirming that inter-VLAN routing via SVIs on Core-SW-HQ is working.

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 10: PC-IT pinging PC-HR — 4/4 replies, 0% loss

Test 2: Cross-Site via GRE Tunnel — PC-IT to PC-Sales

PC-IT (HQ) successfully pings PC-Sales (Branch, 192.168.40.10), confirming end-to-end connectivity through the GRE tunnel.

```
C:\>ping 192.168.40.10

Pinging 192.168.40.10 with 32 bytes of data:

Reply from 192.168.40.10: bytes=32 time<1ms TTL=125
Reply from 192.168.40.10: bytes=32 time<1ms TTL=125
Reply from 192.168.40.10: bytes=32 time<1ms TTL=125
Reply from 192.168.40.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.40.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 11: PC-IT pinging PC-Sales across the GRE tunnel — 4/4 replies

Test 3: Traceroute — PC-IT to PC-Sales

The traceroute confirms the traffic path: Core-SW-HQ (192.168.10.1) to Router-HQ (192.168.1.1) to Router-Branch via GRE tunnel (172.16.0.2) to PC-Sales (192.168.40.10). Four hops total.

```
C:\>tracert 192.168.40.10

Tracing route to 192.168.40.10 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.10.1
  2   0 ms      0 ms      0 ms      192.168.1.1
  3   0 ms      0 ms      0 ms      172.16.0.2
  4   0 ms      0 ms      0 ms      192.168.40.10

Trace complete.
```

Figure 12: Traceroute showing 4 hops through GRE tunnel (172.16.0.2)

Test 4: ACL — Branch to IT (Allowed)

PC-Sales successfully pings PC-IT (192.168.10.10), confirming that Branch-to-IT traffic is permitted by ACL 101.

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=125
Reply from 192.168.10.10: bytes=32 time<1ms TTL=125
Reply from 192.168.10.10: bytes=32 time<1ms TTL=125
Reply from 192.168.10.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 13: PC-Sales pinging PC-IT — ALLOWED (4/4 replies)

Test 5: ACL — Branch to HR (Blocked)

PC-Sales cannot reach PC-HR (192.168.20.10). The ACL deny rule is actively blocking this traffic, as confirmed by the 'Destination host unreachable' responses.

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 14: PC-Sales pinging PC-HR — BLOCKED (100% loss)

Test 6: ACL — Branch to Management (Blocked)

PC-Sales cannot reach PC-Admin (192.168.30.10). The ACL deny rule blocks Branch access to the Management VLAN, protecting admin resources.

```
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Figure 15: PC-Sales pinging PC-Admin — BLOCKED (100% loss)

10. Lessons Learned

Layer 3 Switch Configuration: The 'ip routing' command must be enabled on the 3560 for SVIs to route traffic. The 'no switchport' command is required on the uplink port to make it a routed interface.

Trunk Encapsulation: On Cisco 3560 switches, you must explicitly set 'switchport trunk encapsulation dot1q' before setting trunk mode, otherwise the command is rejected.

GRE Tunnel Design: GRE tunnels require the underlying WAN route to remain reachable independently of the tunnel routes. Routing tunnel traffic through the tunnel itself creates a recursive routing loop.

Static Route Next-Hop: Router-HQ serves as the central routing hub. It must have specific routes to all subnets, while spoke devices (Core-SW-HQ, Router-Branch) can use default routes pointing to Router-HQ.

ACL Placement: Applying ACLs inbound on the source interface (Router-Branch GigE0/0) blocks unwanted traffic as early as possible, reducing unnecessary processing across the network.

Router Selection for Labs: Not all Packet Tracer router models support the same feature sets. The 2911 was used because the 1941 did not support GRE tunnel interfaces in this Packet Tracer version.

End of Documentation

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